PRODUCT REVIEW SENTIMENT ANALYSIS

Assignment 1: Data Acquisition, Text Processing and Cleaning

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Course	Introduction to Natural Language Processing
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This report documents dataset inspection, exploratory analysis, tokenization experiments and a reusable text-cleaning pipeline for Phase 1 of the NLP semester project.

Submission reminder: The course brief requires a registered team of 2–5 students and a contribution statement. Add actual team-member names and their real contributions before submission. Do not claim work by members who did not contribute.

1. Introduction

The purpose of this phase is to acquire a public text dataset, inspect its structure, explore the review text and prepare a reproducible cleaning process. The selected topic is product review sentiment analysis. Customer reviews contain opinions about products and can be grouped into sentiment categories.

2. Dataset Description

The project uses the Women's E-Commerce Clothing Reviews dataset hosted on Kaggle. The dataset file used is *Womens Clothing E-Commerce Reviews.csv*. The text field used for analysis is *Review Text*, and the numeric *Rating* field is used to derive three sentiment classes. The original CSV is stored locally in *data/raw/* and is not committed to Git.

Dataset source: Kaggle — Women's E-Commerce Clothing Reviews. Verify the current license shown on the dataset page and record it in the final README.

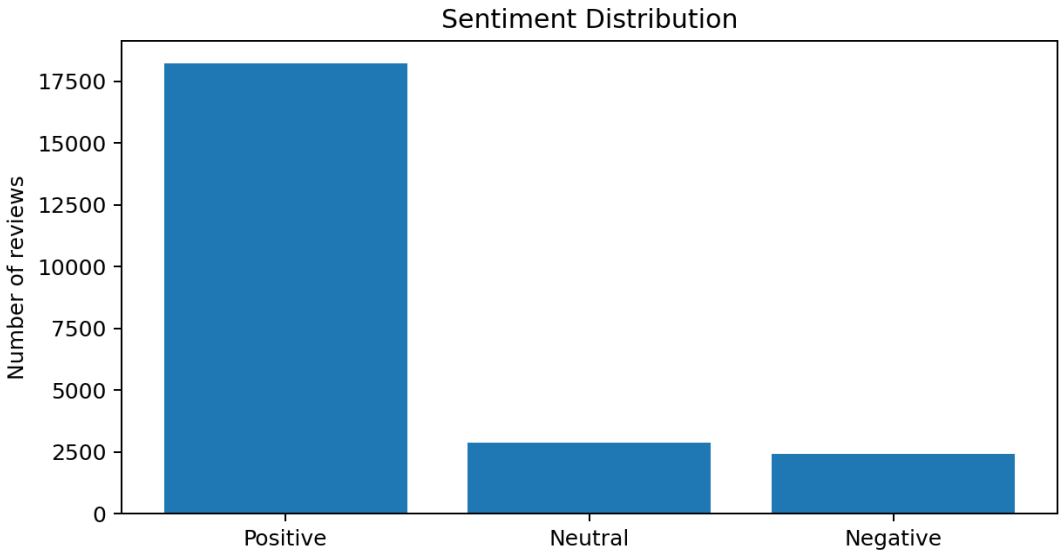
3. Dataset Exploration

Measure	Observed result
Dataset shape	23,486 rows × 12 columns
Average review length	60.2 words
Sample token count	61,962 tokens
Sample vocabulary (types)	3,667 unique types

The token and type counts were calculated on the 1,000-review sample used in the notebook. A token is each word occurrence, including repetitions; a type is a distinct word form. The full dataset shape reported by the notebook is 23,486 records and 12 columns.

4. Sentiment Label Creation

The Rating field was converted into a three-class sentiment label. Ratings 1–2 were assigned Negative, rating 3 was assigned Neutral, and ratings 4–5 were assigned Positive. These are derived labels created for this project, not original three-class dataset annotations.



Sentiment	Reviews
Positive	18,208
Neutral	2,871
Negative	2,407
Total	23,486

5. Tokenization Methods

Regex tokenizer: The notebook uses the pattern *\b\w+\b* after converting text to lowercase. It extracts word-like units and generally excludes punctuation.

NLTK tokenizer: NLTK's *word_tokenize()* was applied to a review. It can separate punctuation into individual tokens and may therefore produce a different token sequence and count compared with the regex method.

The comparison demonstrates that tokenization choice affects how punctuation, contractions and hyphenated expressions are represented. The notebook contains the example output.

6. Useful Regular Expressions

Purpose	Pattern	Explanation
URL removal	https?://\S+|www\.\S+	Matches common http/https links and www links.
HTML tag removal	<[^>]+>	Matches text enclosed in angle brackets, such as HTML tags.
Whitespace normalization	\s+	Matches one or more whitespace characters so they can be replaced by one sp

7. Dataset-Specific Tokenization Challenges

1. **Punctuation:** Customer reviews can contain commas, full stops, exclamation marks and question marks. Different tokenizers may retain or separate these marks differently.

2. **Contractions:** Forms such as “don't” and “isn't” can be split into more than one token. This matters because negative meaning should be preserved during preprocessing.

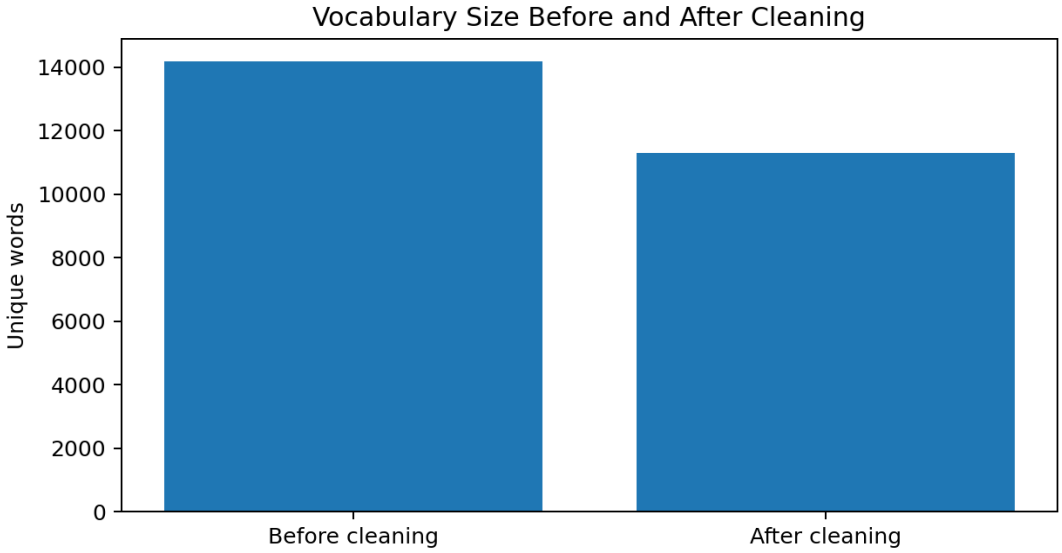
3. **Hyphenated words:** A tokenizer may keep a hyphenated expression as one unit or split it into separate words.

8. Text Cleaning Pipeline

A reusable function was implemented in *src/clean_data.py*. It handles missing review values, applies Unicode NFC normalization, converts text to lowercase, removes URLs and HTML tags, extracts alphanumeric word tokens, removes English stopwords and applies WordNet lemmatization. The function is applied to the Review Text column and the processed dataset is saved as *data/processed/cleaned_reviews.csv*.

Negation handling: The stopwords “not”, “no” and “nor” are excluded from the stopword set so these important negative terms are retained. Removing negation can reverse or weaken a review's meaning.

9. Vocabulary Reduction



Measure	Value
Vocabulary before cleaning	14,183
Vocabulary after cleaning	11,312
Vocabulary reduction	2,871
Reduction percentage	20.25%

The vocabulary decreased by 2,871 unique words (approximately 20.25%) after cleaning. This indicates that normalization, noise removal, stopword filtering and lemmatization reduced variation in the text. The original reviews remain unchanged; cleaned reviews are stored in a separate processed CSV file.

10. Zipf's Law and Long-Tail Pattern

The notebook plots word frequencies in descending order on logarithmic axes. In natural language, a small number of words usually occur very frequently, while a large number occur rarely. This creates a long-tail frequency distribution commonly associated with Zipf's Law. The notebook graph and printed top-word frequency list provide the visual and numerical evidence.

11. Reproducibility and Repository

The project uses a Python virtual environment and dependencies listed in *requirements.txt*. The repository includes a data-loading script, an exploration notebook and a reusable cleaning script. The raw CSV is excluded from version control through ignore rules. A user must download the dataset from Kaggle and place it in *data/raw/* before running the scripts.

Commands:

```
python -m venv .venv
pip install -r requirements.txt
python src/load_data.py
python src/clean_data.py
```

12. Conclusion

In Phase 1, a public product-review dataset was selected and inspected. Exploratory statistics were calculated, sentiment labels were derived from ratings, two tokenization approaches were tested, useful regex patterns were documented, and a reusable text-cleaning pipeline was created. The processed CSV is prepared for the next phase, where numerical text representations and machine-learning methods can be explored.

Before submission: Confirm the dataset license on Kaggle, add real team-member names and contributions required by the course brief, and ensure the notebook and scripts run from a fresh clone. Do not invent team contributions.